Pattern Recognition: Homework 2

Due date: 2026.3.9

Problem 1 (20 pt)

Suppose there is a linear model

$$f_{\boldsymbol{w},b}(\boldsymbol{x}) = \boldsymbol{w}^\top \boldsymbol{x} + b,$$

where $\boldsymbol{w} \in \mathbb{R}^d$ and $b \in \mathbb{R}$ are model parameters. Stack all n input vectors as the matrix $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n] \in \mathbb{R}^{d\times n}$, where each column $\boldsymbol{x}_i \in \mathbb{R}^d$ corresponds to one input sample. Stack all target outputs as the vector $\boldsymbol{y} = [y_1, y_2, \ldots, y_n]^\top \in \mathbb{R}^n$. Let $\boldsymbol{w}^*, b^*$ be the optimal solution that minimizes the sum of squared errors $\sum_{i=1}^n \left(f_{\boldsymbol{w},b}(\boldsymbol{x}_i) - y_i\right)^2$. Denote the predicted value vector as $\hat{\boldsymbol{y}} = \boldsymbol{X}^\top \boldsymbol{w}^* + b \cdot \mathbf{1}$, where $\mathbf{1} \in \mathbb{R}^n$ is an n-dimensional all-one vector, and the residual vector as $\boldsymbol{e} = \boldsymbol{y} - \hat{\boldsymbol{y}}$. Prove the following two core properties:

- The sum of all residuals is zero, i.e., $\sum_{i=1}^n e_i = 0$;
- The residual vector is orthogonal to the feature vectors, i.e., $\boldsymbol{X}\boldsymbol{e} = \mathbf{0}$, and the residual vector is also orthogonal to the fitted value vector, i.e., $\hat{\boldsymbol{y}}^\top \boldsymbol{e} = 0$.

Problem 2 (30 pt)

Given the sigmoid function defined as $\sigma(z) = \frac{1}{1+e^{-z}}$, the logistic regression model $p(y=1|\boldsymbol{x}) = \sigma(\boldsymbol{w}^\top \boldsymbol{x} + b)$ with labels $y \in \{0, 1\}$, and the cross-entropy loss $L = -y \log p - (1-y)\log(1-p), p = p(y=1|\boldsymbol{x})$. Complete the following calculations and round your results to 4 decimal places:

- Given a single sample with feature $\boldsymbol{x} = [1, 2]^T$, weight $\boldsymbol{w} = [0.5, -0.3]^T$, and bias $b = 0.2$, calculate the probability $p(y=1|\boldsymbol{x})$ of this sample belonging to the positive class.
- If the true label of this sample is $y = 1$, calculate the cross-entropy loss value of this sample.
- Calculate the gradient of the loss with respect to the weight $\boldsymbol{w}$ ($\frac{\partial L}{\partial \boldsymbol{w}}$) and the gradient with respect to the bias b ($\frac{\partial L}{\partial b}$).

Problem 3 (10 pt)

For both logistic regression and single-layer perceptron, justify why they are classified as linear models, despite the inclusion of non-linear activation functions.

Problem 4 (40 pt)

In this problem, you need to write a linear classifier in different ways to get a taste of the content in class. **Please notice that you should not use any package that solves the problem in very few lines like `scipy.stats.linregress`. You should only use package like `numpy` to build up the model on your own, otherwise you will not get any points.**

You will write a classifier for predicting whether a person is likely to have breast cancer. In the attachment is our data file `breast-cancer-wisconsin.txt`. The file consists of 699 lines, each line with 11 integer attributes (or features) as below

Attribute Domain

```
1.  Sample code number            id number
2.  Clump Thickness               1 - 10
3.  Uniformity of Cell Size       1 - 10
4.  Uniformity of Cell Shape      1 - 10
5.  Marginal Adhesion             1 - 10
6.  Single Epithelial Cell Size   1 - 10
7.  Bare Nuclei                   1 - 10
8.  Bland Chromatin               1 - 10
9.  Normal Nucleoli               1 - 10
10. Mitoses                       1 - 10
11. Class:                        (0 for benign, 1 for malignant)
```

1 (10 pt)

Adopt Fisher's criterion to find the optimal linear classifier using attributes 2 to 10 to predict label 11. Give the 9-dimensional unit norm vector for $\boldsymbol{w}^*$, and the classification accuracy on the dataset.

2 (20 pt)

Using logistic regression, namely the classifier $f(\boldsymbol{x}) = \sigma(\boldsymbol{w}^\top \boldsymbol{x} + b)$ in problem3 to do the classification. You can follow the procedure below

- 1. Randomly sample the initial parameter $\boldsymbol{w}_0$ from i.i.d. Gaussian and choose $b_0 = 0$.
- 2. Use loss function $L(\boldsymbol{w}, b) = \sum_{i=1}^{N} \frac{1}{2}(\sigma(\boldsymbol{w}^\top \boldsymbol{x}_i + b) - y_i)^2$ to compute the loss value of the current classifier on all the 699 data.
- 3. Compute the gradient $\left.\frac{\partial L}{\partial \boldsymbol{w}}\right|_{\boldsymbol{w}_t}, \left.\frac{\partial L}{\partial b}\right|_{b_t}$.
- 4. Pick a proper (small) value ρ, update $\boldsymbol{w}_t = \boldsymbol{w}_{t-1} - \rho \nabla_{\boldsymbol{w}} L(\boldsymbol{w}, b)$, $b_t = b_{t-1} - \rho \nabla_b L(\boldsymbol{w}, b)$.
- 5. Go back to 2 until the loss is sufficiently low (or repeat for enough iterations).

You need to specify the ρ you use, plot the loss value against iterations, and report the final classification accuracy.

Problem 4 (40 pt) continued on next page. . .

3 (10 pt)

Compare the cosine between two $\boldsymbol{w}^*$ you get in sections 1 and 2. How similar are they? Why?
And try to figure out the most indicative feature that implies one gets breast cancer from $\boldsymbol{w}^*$.

Problem 5 (20 pt) ★

(★ Required for Qiuzhen students; optional for others.)
In our class, we have learned the deduction for Fisher's criterion as maximum the ratio $J_F(\boldsymbol{w}) = \frac{S_b}{S_w}$. Actually, there is another way to deduce it. Suppose we have a set of points $\{(\boldsymbol{x}_i, y_i)\}, i = 1, \ldots, N$ where $\boldsymbol{x}_i \in \mathbb{R}^d$ and $y_i \in \{\frac{N}{N_1}, -\frac{N}{N_2}\}$. There are N_1 positive sample with positive label value N/N_1 and vice versa, $N_1 + N_2 = N$. We build a classification function as $f(\boldsymbol{x}; \boldsymbol{w}, b) = \boldsymbol{w}^\top \boldsymbol{x} + b$. The classification error for a certain data point is defined as $L(f(\boldsymbol{x}_i), y_i) = \frac{1}{2}(f(\boldsymbol{x}_i) - y_i)^2$. Prove that the Fisher's criterion for selecting $\boldsymbol{w}^*$ in our class as

$$\boldsymbol{w}^* = S_W^{-1}(\boldsymbol{m}_1 - \boldsymbol{m}_2),$$

is parallel to the $\boldsymbol{w}^\star$ in solution

$$\boldsymbol{w}^\star, b^\star = \text{argmin}_{\boldsymbol{w},b} \frac{1}{N} \sum_{i=1}^{N} L(f(\boldsymbol{x}_i; \boldsymbol{w}, b), y_i).$$